Share the Cost of Cleaner Orchard Spraying

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Student proposal, not an approved county program. Budget caps are not quotes. No group endorsement is claimed.

Public proposal: https://bkzzzz.github.io/econ238-portfolio/week-seven-orchards.html

Anonymous ballot:
https://docs.google.com/forms/d/e/1FAIpQLScF0LJ2AFTetdtth8oCl_rHtiA3wnIO1NwrAzXFgJzgWPac3g/viewform?usp=header

The ballot question

Shall Monroe County authorize a two-season, 50-acre orchard precision-spraying pilot, appropriating up to $20,000: $15,000 toward one retrofit and $5,000 for independent evaluation? The grower must match equipment funding, operate the system for five seasons, and share pesticide-use, drift-test, and crop-quality records. No acceptable bid and signed grower agreement means no pilot.

What's the deal?

County taxpayers contribute at most $20,000 from general revenues, displacing other county spending. A competitively selected, consenting grower contributes at least half the installed equipment cost and pays training, maintenance, and operating costs. The equipment allowance is $30,000; the public reimbursement is half the invoice, capped at $15,000. Independent evaluation has a $5,000 ceiling. These are proposed budget caps, not vendor quotes or approved funds. One existing sprayer would serve 50 acres for five seasons, with intensive evaluation in the first two. We buy a chance to reduce off-target pesticide deposition and publish useful local evidence, while retaining apple production. Neighbors and workers may benefit; the grower retains input savings and bears performance risk. Taxpayers bear the subsidy and evaluation costs. If qualified equipment and evaluation bids do not fit, do not proceed.

The best case for YES

The relevant alternative is continued conventional spraying, not an orchard without any pesticides. A retrofit could preserve crop protection while reducing chemicals landing beyond the trees. USDA research establishes that this technology exists; it does not establish the benefits at a particular Rochester-area orchard. A small trial would let us check whether it works locally before considering a wider subsidy. The grower's match and five-season operating obligation give the farmer a stake in making it work. Public reporting can help other growers decide whether to adopt without another grant. I would support the trial if an independent evaluator can measure off-target deposition within the budget. The argument is strongest where nearby people face meaningful exposure and the orchard would otherwise delay adoption.

The best case for NO

Why should taxpayers buy equipment that lowers a private business's chemical bill? A grower expecting large savings could finance the retrofit without a grant. Paying an orchard that would upgrade anyway creates a windfall, not an additional environmental improvement. Taxpayers also surrender the next-best use of $20,000, and people elsewhere may value other projects more. Two seasons of weather cannot establish long-run health effects, while $5,000 may not buy credible independent evaluation. Small changes in crop damage can erase chemical savings. Competitive selection may favor an orchard with good grant-writing skills rather than the largest external harm. A stronger policy might enforce existing protections or target documented exposure directly. Without a credible additionality test and affordable measurement plan, I would vote NO.

Where is the body buried?

The weakest assumption is that the subsidy changes the grower's timing rather than reimbursing an already-profitable purchase. We have no participating orchard, installed-price quote, or evaluator bid. Less spray is not automatically less harm: chemical toxicity, weather, and exposure matter. The model does not establish a health benefit or positive net welfare. Taxpayers can lose money even if the sprayer works; the orchard can lose crop value if it does not. The budget gates must be real, even if that means cancelling the pilot.

Why isn't this already happening?

Some adoption already is happening: USDA reports commercialization in 2019, and Cornell has hosted regional precision-spraying programming. The question is why a particular orchard has not adopted. Capital constraints, uncertain equipment compatibility, maintenance, short planning horizons, and crop-loss risk can delay investment. Neighbors cannot easily bargain with a grower when many households benefit and no one wants to pay the whole bill. Information from one orchard can also benefit competitors who did not finance the trial. But slow adoption may be rational if private savings are small. Before offering money, request dated investment plans, financing alternatives, and a bid for the minimum subsidy needed. Compare equipment-only private financing with a public evidence contract. Exclude committed purchases, disclose other subsidies, and prevent duplicate reimbursement. These screens reduce windfalls; they cannot prove the counterfactual.

Implementation and economic check

Who decides?

This is a proposed county-funded pilot, not a currently authorized ballot measure. County approval, confirmed spending authority, an appropriation, procurement and legal review, a voluntary grower contract, and an independent evaluator are prerequisites. New York already operates agricultural conservation cost-sharing through county Soil and Water Conservation Districts. That is an institutional precedent, not confirmation that this equipment or funding proposal is eligible. The county must establish an authorized contracting or cost-share route before spending. No county, grower, vendor, or Cornell commitment is claimed.

How to deliver it

Seek written bids for an installed retrofit on an existing compatible sprayer and an independent two-season evaluation. Select one 50-acre site by proximity to potential receptors, credible monitoring access, and the smallest necessary subsidy. Do not pay for a purchase already ordered. Execute the grower and evaluation agreements only after both fit the caps; otherwise return the appropriation unspent.

What gets measured

Pre-register paired conventional/precision test runs under comparable weather, canopy and travel speed, using a safe tracer and identical sampling positions beyond the orchard boundary. Record deposition per unit ground area, application rates by product, treated acreage, weather, crop damage and marketable yield. Compare matched blocks in each of the first two seasons. A proposed 20% decrease in off-target tracer deposition is a decision threshold, not a promised effect. Do not substitute liters of tank mix for toxicity or exposure.

Accountability

The farmer funds at least 50% of installed equipment, all recurring costs, and any overruns; the public match remains capped. Require five seasons of operation, inspection access, and annual records. Reimburse documented installation after verification, with a contract providing proportional recovery for early withdrawal or failure to operate. Bad scientific results alone are not misconduct. Publish null results and crop losses. After two seasons, reconsider the contract if crop protection is materially worse; do not automatically expand the subsidy.

The counterfactual

Compare with the same orchard using its current sprayer and with a privately financed upgrade. If the upgrade would happen without the subsidy, count only acceleration and genuinely additional public information. Routine compliance with pesticide rules remains mandatory under either option. This pilot does not purchase permission to violate labels or expose neighbors.

Illustrative five-year calculation

Illustrative assumptions: 50 acres; $30,000 installed equipment; $5,000 evaluation treated conservatively as an upfront cost; $1,000 annual incremental training/maintenance/operation; five years; 3% real discount rate; zero salvage. Annual chemical savings per acre are scenarios, not an orchard quote. The farmer pays $15,000 initially; taxpayers pay $20,000. No yield loss, extra labor savings, or monetized health benefit is assumed. A subsidy is a transfer between payer and recipient; count the full equipment and evaluation resource costs once in the social comparison. Tax-finance distortions and procurement overhead are omitted, so positive modeled values are not proof of a good policy.

PV factor = sum of 1/(1.03)^t for years 1 through 5 = 4.579707. Private NPV = (50s - 1,000) x factor - 15,000. Social NPV before external benefits = (50s - 1,000) x factor - 35,000. Required annual external benefit = max(0, -social NPV)/factor. Here s is annual chemical savings in dollars per acre.

Saving / acre	Gross / year	Grower NPV	Social NPV*	External / year**
$56	$2,800	$-6,757	$-26,757	$5,842
$100	$5,000	$3,319	$-16,681	$3,642
$200	$10,000	$26,217	$6,217	$0

* Before external benefits. ** Annual external benefit needed to close the modeled gap.

At $100 annual chemical savings per acre, private gains can coexist with a social cost gap. If private adoption would occur anyway, the public payment requires an additional justification.

Sources and limitations

USDA ARS: Precision Sprayer Benefits Growers and the Environment

https://www.ars.usda.gov/oc/dof/precision-sprayer-benefits-growers-and-the-environment/

Documents existing retrofits, commercialization, and multi-crop field trials. Reported chemical savings range from $56 to $812 per acre annually across crop types; this is not a local apple-orchard forecast.

Cornell Cooperative Extension: High Tech Precision Orchard Spraying

https://lof.cce.cornell.edu/event.php?id=1440

Shows regional extension involvement and commercial orchard-spraying equipment. Cornell has not agreed to evaluate this proposal.

USDA ARS: Laser-Guided Crop Sprayer (2016)

https://agresearchmag.ars.usda.gov/2016/feb/lasersprayer/

Separately identifies apple-orchard drift tests and nursery cost savings. Nursery savings should not be relabeled as apple savings.

New York State Board of Elections: 2022 environmental bond question

https://results.elections.ny.gov/contest/244

A model for concise authorization language, not legal authority or funding for this pilot.

New York Agriculture and Markets: Agricultural Nonpoint Source Abatement and Control

https://agriculture.ny.gov/node/5336

Confirms that agricultural conservation cost-sharing through county Soil and Water Conservation Districts exists. It does not confirm eligibility or an award for this sprayer pilot.

The October 22 interview is still planned. This proposal does not use interview evidence. The class poll results are reported separately.